

Can a soundtrack's emotion predict a film's genre?

346 clips · 43 films · Eerola & Vuoskoski (2011) Implementation complete. Open item: content chapters.

The three things to say first

1. Emotion beats direct audio on all 8 genres — and it is now statistically significant ($p = 0.031$). Last time this was $p = 0.62$.
2. Audio-predicted emotions work as well as human ratings ($p = 0.855$), so the pipeline needs no emotion annotations.
3. The emotion regression's $R^2 = 0.56$ is 62% of the maximum any model could reach — two listener panels only agree at 0.90.

Where the four research questions stand

RQ	Question	Status
RQ1	Can emotion predict genre?	Yes — 0.397 on 5 genres vs 0.166 chance
RQ2a	Which emotions relate to which genres?	Done — per-genre Cohen's d signatures
RQ2b	Does the emotion intermediate help vs direct audio?	Yes on 8 genres ($p = 0.031$); borderline on the 5-genre subset
RQ3	Which emotions are strongest per genre?	Done — coefficients, RF importance and effect sizes agree

Four numbers changed since last time

Previously reported	Now	Why
Emotion vs AST “within noise”, $p = 0.62$	Significant, $p = 0.031$	5 folds can't resolve that gap; 50 folds + tuning can
0.281 is a “robust ceiling”	Ceiling is 0.299	Regularisation was never tested; tuning alone beats it
Predicted-emotion pipeline = 0.406	0.394	One split was optimistic; 50 folds regress it
338 clips	346 clips	Corrected inclusion rule

Three of the four moved because the **evaluation got stricter**, not because the models improved. That progression — $p = 0.62 \rightarrow 0.092 \rightarrow 0.031$ — is itself a reportable finding about how little one cross-validation run can tell you.

The result

Audio → emotion → genre, against classifying genre straight from audio. 10×5 repeated GroupKFold by film, nested-CV-tuned, Macro-F1.

Five-genre subset (329 clips, 41 films)	Macro-F1	95% CI
Ground-truth emotion — ceiling of the emotion path	0.397	0.388–0.407
VGGish → predicted emotion — the real pipeline	0.394	0.389–0.400
VGGish-128, direct	0.350	0.335–0.365
AST-768, direct	0.345	0.328–0.361
PCA-8 of VGGish — control for “any 8-d bottleneck”	0.343	0.330–0.355
CLAP-512, direct	0.335	0.323–0.347
Most-frequent dummy	0.166	0.165–0.167

On all eight genres: emotion 0.300 vs AST 0.257, dummy 0.101.

Comparison	Δ	p	Folds won	Verdict
Emotion vs AST, 8 genres	+0.043	0.031	92%	Significant
Emotion vs CLAP, 5 genres	+0.062	0.036	86%	Significant
Predicted emotion vs PCA-8 control	+0.052	0.060	88%	Borderline
Predicted emotion vs AST, 5 genres	+0.050	0.078	82%	Borderline
Predicted vs ground-truth emotion	−0.003	0.855	54%	Same

AST vs CLAP	+0.010	0.752	62%	Same
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- **The three pretrained embeddings are indistinguishable from each other** (AST 0.345, VGGish 0.350, CLAP 0.335) and emotion beats all three — so the baseline wasn't just a badly chosen audio model.
- **Predicted emotion = ground-truth emotion.** The pipeline doesn't depend on human ratings, which is what makes it usable at all.

How it is actually built

Stage	What runs	Detail
Audio → vector	Frozen pretrained model, mean-pooled over time	AST 768-d (primary), VGGish 128-d, CLAP 512-d, librosa-MIR 103-d. No fine-tuning anywhere — pure feature extraction, cached per clip.
Stage 1: vector → 8 emotions	RandomForestRegressor, 300 trees	One multi-output forest predicting all 8 emotions jointly. Chosen empirically: Ridge 0.37, SVR-RBF 0.54, RF 0.56 — the regressor mattered more than the features.
Stage 2: 8 → 11 → genre	Binary relevance logistic regression	8 emotions plus valence×energy and a positive/negative composite. StandardScaler → one L2 LogReg per genre, class_weight=balanced, C≈0.003 chosen by nested CV.
Per fold	Everything fit on training clips only	Scaler, PCA, forest and the C value are all fit inside the fold; no test clip touches any model. Seed 42 throughout — re-runs are bit-identical.

The one technical detail worth pre-empting

The genre classifier is trained on the forest's *in-sample* emotion predictions ($R^2 = 0.94$ on training clips) but evaluated on out-of-sample ones ($R^2 = 0.60$). That is a train/test distribution mismatch, *not* leakage — no test clip or label influences either fit, so the score stays an honest generalisation estimate. The direction matters: the classifier is calibrated on clean features and then handed noisy ones, so 0.394 is if anything a slight *under*-estimate. The textbook fix is an inner cross_val_predict so both sides share a distribution. Better to state this than to have it drawn out.

Why the method holds up

Decision	Why it is defensible
Cross-validation grouped by film	AST scores 0.44 ungrouped vs 0.26 grouped — a +0.18 gap. ~40% of its apparent performance was recognising which film a clip came from. Emotion features leak only +0.05.
Macro-F1 as headline metric	Under this imbalance the dummy wins exact-match and Hamming loss. Macro-F1 weights every genre equally, so predicting only Drama scores badly.
50 folds, not 5	Five folds can't separate a 0.05 difference from partition noise. Significance uses the Nadeau–Bengio corrected test, since overlapping training sets break the ordinary t-test.
Regularisation tuned by nested CV	The default C = 1.0 was the worst setting for every feature set. Tuning helped the baselines more than emotion (+0.033 CLAP vs +0.018 emotion), so it doesn't flatter us.
Both 5 and 8 genres reported	Genre count scales with corpus size (Austin: 4 over 98 films; Ma: 6 over 110). Biography (24) and Documentary (17) are too rare; Adventure has no emotional signature at all.
Binary relevance, not chains	Chains were tested and are significantly worse ($p = 0.024$) — rejected empirically, not assumed.
Single 10.24 s window	Full-clip pooling (first/centre/mean/max) was tested; all differences ≤ 0.006 . Short homogeneous cues — the opening 10 s already characterises the clip.

How the pipeline compares with related work

Stage	This thesis	Related work says
Embedding	AST-768 (frozen)	Kang & Herremans (2025) name AST a leading pretrained embedding

		for MER.
Pooling	Mean-pool over time	Behrouzi (2023) mean-pools MFCC per section; Ma (2021) resamples VGGish frames to 1 Hz.
Multi-label	Binary relevance	Mangolin (2020) uses BR + ML-kNN; Behrouzi (2023) explicitly corroborates BR with a discriminative classifier.
Metric	Macro-F1	Ma (2021), the closest comparator, also reports macro-F1.
Emotion stage	Audio → emotion → genre	None of the four genre studies has one — this is the contribution.
Grouped CV	GroupKFold by film	Only Ma (2021) gets it implicitly, via film-level MIL bags. Austin, Mangolin and Behrouzi do not guard against leakage.
Significance	Repeated CV + corrected t-test	Not done anywhere in the reviewed literature — above field norm.

Two things to raise before they are asked

Emotion R^2 against the MER benchmark. Kang & Herremans report $R^2 = 0.55$ for valence and 0.79 for arousal. Ours: valence 0.616 (above), tension 0.664 and energy 0.612 (below). We beat the field on the hard dimension and trail it on the easy one — different datasets, so positioning rather than a head-to-head. **Our Blockbuster result diverges from Ma et al.** They report VGGish not substantially beating hand-crafted MIR; we find VGGish 0.582 vs MFCC 0.455. But we use 78 of their 140 MIR features (MFCC family only) and mean-pool cues instead of MIL attention pooling — so this is most likely the reduced feature set and simpler pooling, not a refutation. State the caveat first.

If they compare our 0.397 with Ma's 0.65

Not comparable, for three reasons. (1) Ma classifies films, we classify clips — our own film-level Blockbuster run reaches 0.582, much closer to theirs. (2) Different corpus: 110 blockbusters with a Score Stamper filter vs 43 films of curated excerpts. (3) Our evaluation is leakage-safe and theirs largely is not — part of the apparent deficit is a deliberate methodological choice, and our own +0.18 leakage measurement shows its size.

How good could any model possibly be?

Set 2 re-rates 110 of the same excerpts with a **different listener panel**, which gives the one thing Set 1 alone cannot: how much of the emotion signal is real.

Quantity	Value	Reading
Between-panel agreement	$r = 0.912$	Two panels, same excerpts
Within-panel repeat trials	$r = 0.988$	Same panel, excerpt shown twice
Ceiling on R^2 this implies	0.897	The best any predictor could reach
Achieved R^2 (random forest)	0.560	62% of attainable variance

Per emotion the spread is wide: tension 0.664, anger 0.630, valence 0.616 are easy; **sad 0.437** and **happy 0.314** are hard — the classic “valence problem”, documented by both Kim (2010) and Kang & Herremans (2025). Our valence (0.616) actually beats their reported 0.55 benchmark.

- **$R^2 \approx 1.0$ was never achievable** — read 0.56 against a ceiling of 0.90, not 1.0.
- **Rating noise does not excuse the gap** — a third of attainable variance is still unexplained, so the limit is the audio representation and corpus size, not the labels.

Which emotions mark which genres

Genre	n	Signature	d	Genre	n	Signature	d
Horror	28	high fear	+0.75	Documentary	17	low anger	-0.51
Biography	24	high happy	+0.67	Action	103	low happy / tension	-0.50
Comedy	40	low fear	-0.61	Crime	98	high fear / low sad	+0.40
Drama	239	high valence	+0.57	Adventure	97	<i>none</i>	± 0.25

Fear is the master discriminator (threat genres vs light genres), valence the second axis. Adventure having no signature *explains* its near-chance predictability rather than leaving it unexplained.

Fix before submission

chapter/fundamentals.tex gives Horror's anger effect as $d \approx 0.51$ and fear as 0.77. The correct values are 0.34 and 0.75 — which makes the argument there *stronger*, not weaker.

Questions to expect

Is the emotion advantage real, or noise?

Significant on 8 genres ($p = 0.031$, 92% of folds). On the 5-genre subset the margins are consistent (80–88% of folds) but $p = 0.060$ – 0.104 , so it is reported as borderline, deliberately.

Isn't emotion just a generic 8-d bottleneck? Any compression would help.

That control was run: PCA-8 of the same embedding scores 0.343 vs the emotion pipeline's 0.394, winning 88% of folds. So it is not merely dimensionality reduction — but at $p = 0.060$ it stays borderline.

Why is the emotion regression only $R^2 = 0.56$?

Because 1.0 was never attainable. Two listener panels agree at 0.897, so 0.56 is 62% of the ceiling. Label noise is therefore not what limits it.

Did you tune, or use defaults?

Tuned by nested CV, inner loop also grouped by film. The default $C = 1.0$ was the worst setting for every feature set, and tuning helped the baselines more than emotion.

Why not an end-to-end neural network?

346 clips from 43 films, with learning curves still rising at 100% of the data — already size-limited for linear models. Gradient boosting was tested and is the worst arm (0.198, $p < 0.001$).

Have you tried harder to improve the score?

Four angles: feature fusion, PCA fusion, gradient boosting, classifier chains, threshold and class-weight changes. Every alternative is below the baseline and wins at most 16% of folds.

Does it transfer beyond this dataset?

Partly. Applied to Ma et al.'s Blockbuster corpus the predicted emotions are face-valid (horror $\rightarrow$ fear +0.99, romance $\rightarrow$ tenderness +1.05). On genre accuracy it ties the control there rather than beating it.

Would more data fix the borderline results?

Yes — and it is provable. The corrected test has a floor that ten repeats already sit within 0.02 of, so the constraint is the 41 films, not the number of folds.

Limitations, and what to ask

- **43 films** is the binding constraint on every borderline result — grouping by film means the effective sample size is films, not clips.
- **The 0.397 figure uses human ratings** — it is the ceiling of the emotion path. The honest end-to-end number is 0.394, which matches it.
- **Biography (17) and Documentary (24) are unlearnable** at this scale, and Adventure has no signature. That is why the 5-genre subset exists.
- **Blockbuster has no ground-truth emotions** — transferred predictions can be checked for face validity but not scored.

Three things worth deciding together:

1. **Is borderline good enough?** Report emotion-vs-control as a consistent but non-significant advantage, or lead with the 8-genre result that is significant? Your call on framing.
2. **Extend the corpus?** Every remaining uncertainty is corpus-bound — the only lever left, though likely out of scope.
3. **Which chapter to draft next?** Methods is still a stub and all the material exists; agreeing its structure makes the writing mechanical.

Status: 5 of 6 supervisor items done; content chapters (#5) is the open one. 17 experiments, reproducible from a clone, with every per-fold score committed. Five thesis-ready LaTeX sections written and mutually consistent.

Terms, if they come up

Term	Meaning	Term	Meaning
Macro-F1	Mean F1 across genres, each weighted equally	Nested CV	Hyperparameters picked inside each training fold
GroupKFold	All clips of one film stay together in a split	Nadeau–Bengio	t-test corrected for folds sharing training data
Binary relevance	One yes/no classifier per genre	ICC	Rater agreement: ranking vs identical values
Cohen’s d	Standardised gap between two means (0.5 ≈ moderate)	AST / VGGish / CLAP	Pretrained audio models: 768-, 128-, 512-d